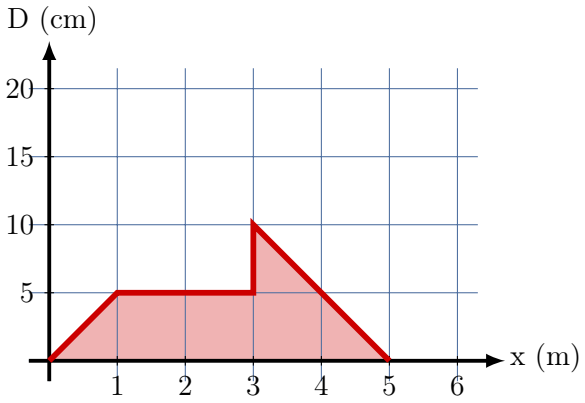
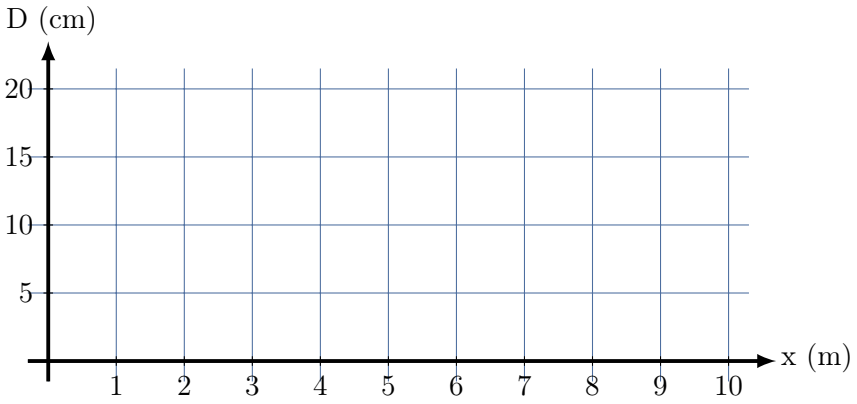
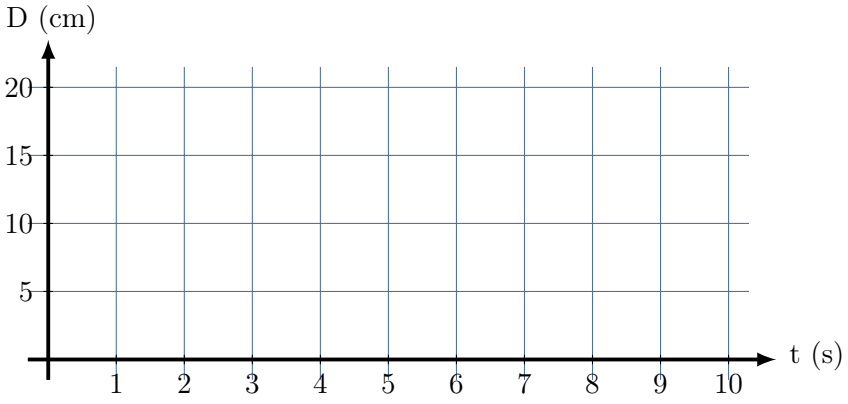
1. The following graph shows the inital **wave pulse** for a wave that is traveling to the right with a velocity of 1 m/s.



A. Draw a **Snapshot Graph** of the wave pulse at t= 3 seconds, D(x,t=3)

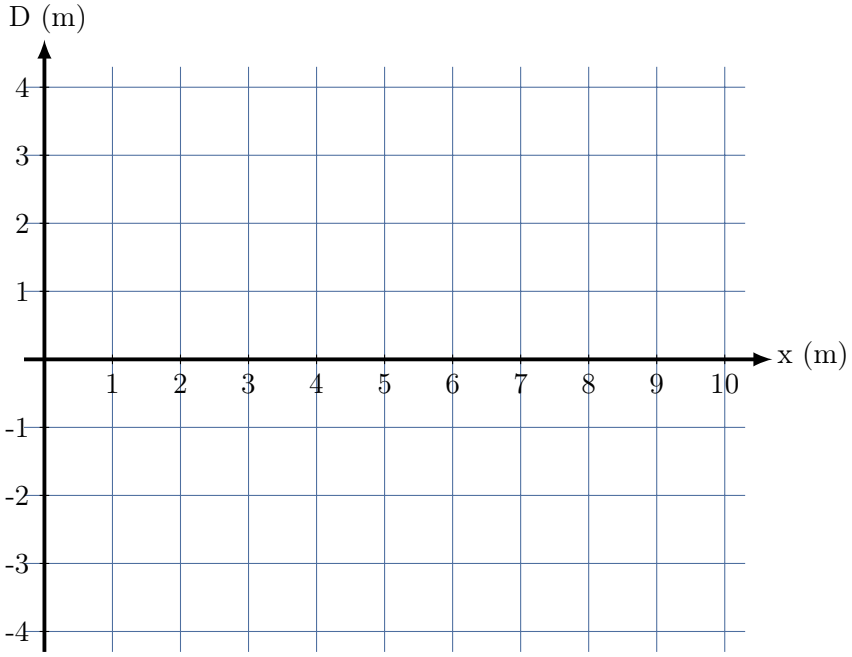


B. Draw a **History Graph** of the wave pulse at the location x = 8, D(x=8,t)



2. Sketch the following wave

$$D(x,0) = 4\sin\left(\frac{2\pi}{5}x + \pi\right)$$



A. What is the wavelength of the wave, λ?

B. What is the wavenumber, k of the wave? What physically does k mean?

C. What is the initial phase of the wave, ϕ_0?